# Latent Modeling of Unexpectedness for Recommendations

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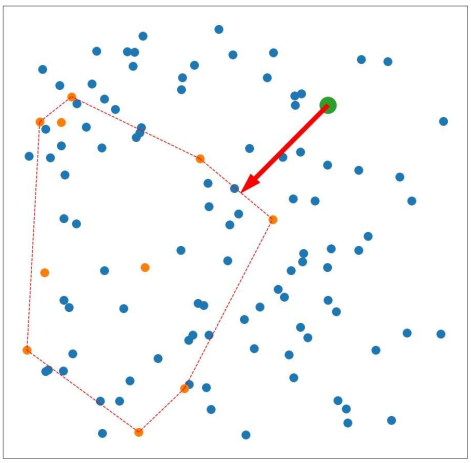
## Problem

- **Traditional recommender systems suffer from the problem of boredom and filter bubbles.**
- **Unexpected recommendation models usually sacrifice performance accuracy to improve novelty.**
- **Explicit modeling of unexpectedness overlooks latent, complex and heterogeneous interactions between users.**

## Solution: Latent Closure (LC) model

- **Model unexpectedness in the latent space**
- **Address user interactions using heterogeneous information network embeddings**
- **Deploy hybrid utilization as mixture of usefulness and unexpectedness**

## Model



**Latent Unexpectedness Measure:**

$$Unexp_{u,i} = d(i; LC(N_i))$$

where i represents user embeddings, LC-Latent Convex Hull and $N_i$-all neighbor embeddings of the user.

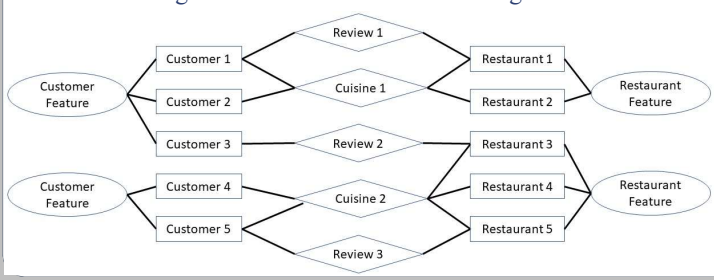
**Hybrid Utility Function:**

$$Utility_{u,i} = (1-\alpha) * EstRating_{u,i} + \alpha * Unexp_{u,i}$$

where *EstRating* represents estimated ratings using classical recommender models and $\alpha$ represents the importance of unexpectedness.

**Heterogeneous Information Network (HIN):**

We use HIN to generate user and item embeddings.



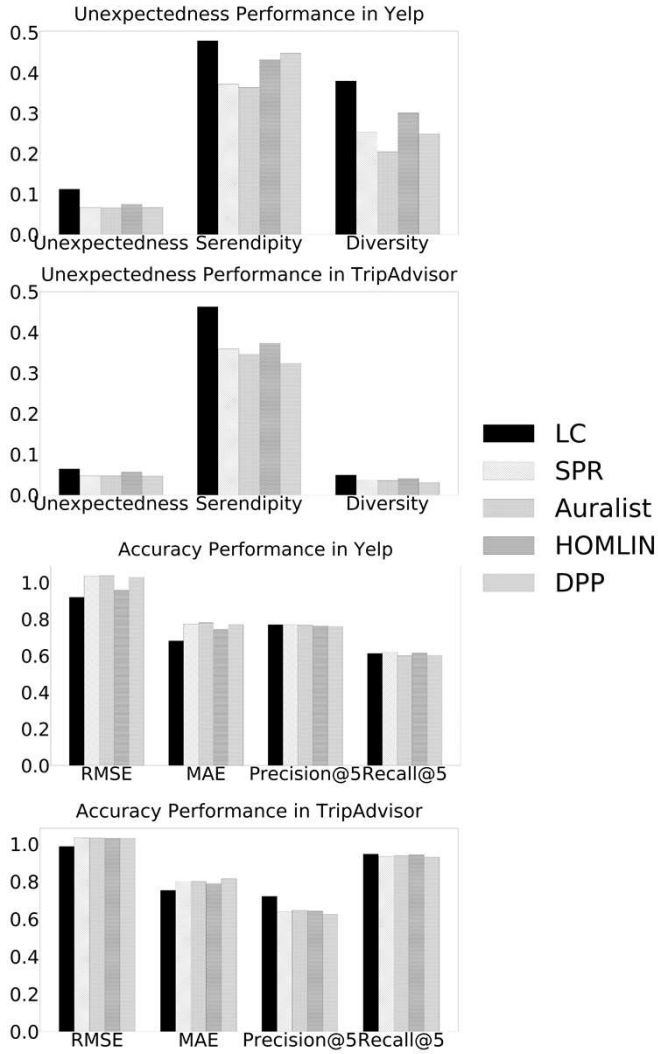

## Experiments

**Datasets**

- Yelp Challenge Round 12
- TripAdvisor Dataset

**Baselines (compared with our LC model)**

- Serendipitous Personalized Ranking (SPR)
- Auralist
- HOM-LIN
- Determinantal Point Process (DPP)




## Conclusions

- **Latent modeling of unexpectedness significantly outperforms baseline models in terms of unexpectedness measures while achieving the same level of accuracy.**
- **The proposed definition of unexpectedness enables us to provide unexpected and useful recommendations simultaneously.**
- **Feature information contributes to the performance of unexpected recommendations, as we achieve greater improvements on Yelp dataset, which contains richer feature information compared to the TripAdvisor dataset.**